Quiz 01

Course: Artificial intelligence (AL2002)

Department of Computer Science

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Section CS-4G

Question 1 — Student Performance Prediction (Regression)

A university wants to predict a student's **Final Exam Score** based on the following features:

- Study Hours per Week
- Attendance Percentage
- Assignment Score
- Class Participation Score

The dataset is given below:

```
data={
"Study_Hours": [2,3,4,5,6,7,8,2,9,5,4,6,7,8,3,5,9,10,1,6],

"Attendance": [60,65,70,75,80,85,90,55,95,78,72,82,88,91,67,76,97,99,50,84],

"Assignment_Score": [55,58,62,68,72,78,85,50,92,70,65,74,80,88,60,71,95,98,45,79],

"Participation": [4,5,5,6,7,8,9,3,10,6,5,7,8,9,4,6,10,10,2,8],

"Final_Score": [50,54,60,66,72,78,86,48,94,69,63,75,82,89,56,70,96,99,42,80]
}
```

Tasks:

- Create a DataFrame using the given dataset.
- Split the data into training and testing sets (80–20).
- Apply feature scaling using StandardScaler.
- Train a **Linear Regression** model.
- Make predictions on test data.
- Evaluate the model using:
  - MSE (Mean Squared Error)
  - RMSE (Root Mean Squared Error)
  - R^2 Score

Question 2 — PyTorch Multi-Class Classification

Use PyTorch to build a neural network model for multi-class classification.

Tasks:

Create a random dataset with:

1200 samples

4 input features

3 output classes

- Convert the dataset into PyTorch tensors.
- Build a neural network containing:
  - Input layer → 4 neurons
  - One hidden layer → 8 neurons
  - Output layer → 3 neurons
- Use **ReLU** activation in the hidden layer.
- Use **Softmax/CrossEntropyLoss** for classification.
- Define the loss function using CrossEntropyLoss.
- Train the model for 60 epochs.
- Print:
  - Epoch number
  - Loss value after each epoch
- Display final prediction results on sample test data.

Exercise 3: Customer Segmentation using K-Means Clustering

Write a Python program to perform customer segmentation using the **K-Means Clustering** algorithm.

Requirements:

- Use make_blobs to generate dataset:
  - n_samples=400
  - centers=4
  - cluster_std=1.2
  - random_state=21
- Apply K-Means clustering:
  - n_clusters=4
- Extract:
  - Cluster labels
  - Cluster centers
  - Inertia value
- Plot:
  - Data points with different cluster colors
  - Centroids using black star (*) marker
- Add:
  - Plot title
  - X-axis label
  - Y-axis label
  - Grid
- Save the plot as:
  - customer_segmentation.png